

Steven William Roche

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EDUCATION

MIT-WHOI Joint Program

PhD Student in Robotics

Cambridge, MA

Expected May 2030

- **GPA:** 5.0 / 5.0
- DoD SMART Scholar and WHOI OVF Fund Recipient

Boston College, Morrissey College of Arts and Science

B.S. in Computer Science, B.S. in Mathematics

Chestnut Hill, MA

May 2025

- **GPA:** 3.96 / 4.00, **SAT:** 1580 / 1600
- **Gabelli Presidential Scholar:** Received full-time merit scholarship awarded to 23 of 40,000 applicants
- **NCA Cheerleader 2022-2024.** Competed in NCA College Nationals twice

RESEARCH EXPERIENCE

MIT, Aerospace Controls Lab and WHOI's WARPLAB

PhD Student

Cambridge, MA & Falmouth, MA

June 2025 – Present

- Used computational fluid dynamics and reinforcement learning to improve the control of an underwater autonomous vehicle. Conducted extensive tank and field testing at a coral reef in USVI. "CORL-AUV" is under review at ICRA
- Extending e-processes to detect & repair tracker failure on vision tracking systems (SAM3, Cutie, etc). Contributing the first maritime segmentation multi-object tracking dataset. Targeting CVPR 2027 with this work
- Co-Chief of Engineering and Control Team Lead, MIT Driverless. Passed A2RL's sim gateways and conducted negotiations with A2RL leadership
- Integration Lead on MIT team working on the NASA Mars to Table Challenge. Worked across teams to research pollination methods, create meals for astronauts, and develop a tradespace pareto front. Semi-finalists in the challenge

NOAA, Cooperative Institute for Research in Environmental Sciences

Research Assistant – Magnetic Navigation Team

Boulder, CO

Jan 2023 – Present

- Architected first use of ML that predicts aircraft location using magnetic readings and no initial position information, which was pitched to Office of Naval Research
- Presented at NOAA Symposium; first author for poster presented at AGU 2024 conference

Boston College, Scholar of the College Thesis

Thesis Writer

Chestnut Hill, MA

June 2024 – Present

- Crafted algorithm that clusters graph vertices (graph theory), minimizing the L-infinity norm on the graph
- Beat the SOTA 4-approximation bound with a 3-approximation bound. Co-author on paper published in ESA 2025

MIT, Brain and Cognitive Sciences Department

Research Assistant – Synthetic Neurobiology Group

Cambridge, MA

May 2022 – Aug 2024

- Collaborated on 2 fish brain analysis projects; contributing author on paper published in Nature Methods
- Optimized 2000+ lines of code; implemented parallel processing, automated 1 ML pipeline, tested & enhanced 5 scripts that register (stitch/align) fish brain images; delivered 2 software packages with corresponding user interface
- Annotated 60+ fish brain layers with 2,000+ neurons per layer; aided training of Cellpose ML annotation model

Oxford University, Mathematical Institute

Visiting Undergraduate Researcher

Oxford, UK

May 2023 – March 2024

- Summarized 20+ academic papers on flood research to predict floods in 30+ large rivers across the globe
- Developed 4 feature LSTM time series & ARIMA models. Tested 31 rivers and compared to physics models

AWARDS

Awards: DoD SMART Scholar (2025), WHOI OVF Research Award (2026), AMS Orville Endowed Scholarship, NOAA Ernest F. Hollings Scholar, BC Sophomore Scholar (top 5%), Advance Study Grant (2022, 2023), General Winner of the BC Hackathon (2021), General Winner of MIT Blueprint Hackathon (2021), Theodore Roosevelt Youth Medal by the US Navy League (2021), Chief in Sea Cadets (2021), Eagle Scout (2019).